The Optimizer Undoes It: Input-Affine Conditioning of a Selective Scan Is Gauge-Absorbable

A Training-Dynamics Null with a Mechanistic Explanation

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Abstract

Selective state-space models (Mamba) have reshaped long-context sequence modeling, yet *which* forms of conditioning survive their training dynamics is poorly understood. We study FiLM-style regime conditioning of a Mamba backbone—a per-block, channel-wise affine on each block's input, meant to steer the input-dependent selective-scan parameters $\Delta, \mathbf{B}, \mathbf{C}$—and find it a **decisive null**: under joint training the modulation collapses to identity and cannot be forced active even by direct supervision. We explain this mechanistically. An input-side affine is a *gauge* direction that folds losslessly into the block's own $\Delta/\mathbf{B}/\mathbf{C}$ projections, so the optimizer slides it back to identity. The signature is dataset- and architecture-invariant: forced-active, efficiency-ratio-supervised escalations decay to identity with a shared time constant ($\tau \approx 7000$ steps, $R^2 > 0.999$) across three very different distributions—FI-2010 equities, Kaggle crypto spot books, and Polymarket prediction markets—implicating the optimizer, not the data. We then sharpen the mechanism with an output-side placement that gates each block's output instead of its input: its multiplicative scale merely folds into the block's *other* (bias-free output) projection and decays with the same constant, while the one genuinely non-absorbable component—an additive shift—decays in lockstep. The null is thus **overdetermined**: gauge absorption removes every direction the host weights can represent, and reconstruction dominance removes the one they cannot. Escaping both would require modulating the scan parameters *inside* the fused kernel, which our hardware cannot run. We demonstrate this boundary at toy scale on the reference scan—an in-kernel modulation is non-absorbable and survives the training that drives the input-affine form to identity—and bound the null with a language-model control: a Mamba char-LM on a strongly-decodable regime (code vs. prose) *does* engage the modulation, so the collapse is specifically the fate of *weakly-decodable*, drop-in regime conditioning. The financial datasets are a demonstration substrate; consistent with a result about how selective scans *train* rather than which is best, a matched-parameter ablation shows no backbone dominates, and the one economic signal we find is reproduced by a logistic regression on the same features—architecture-independent, as the null predicts.

1 Introduction

A practitioner who wants a sequence model to behave differently in different regimes will reach for the most natural tool available: condition the model on an inferred regime. For a Mamba backbone the most kernel-friendly way to do this is a FiLM-style affine on each block's input, because the selective scan's parameters $\Delta_t, \mathbf{B}_t, \mathbf{C}_t$ are themselves input-dependent (§3)—so an affine on the input propagates a regime signal *into* the dynamics without touching the fused CUDA kernel. It is the drop-in form of the idea. **We find that it does not survive training, and we explain why.**

We attach an exactly zero-initialized FiLM modulator (a regime router plus a hypernetwork emitting per-block γ, β) to a Mamba-3 world model and train it jointly with the host objective.

Despite the cleanest possible control—at initialization the modulated backbone is *bit-identical* to the unmodulated one—and despite the strongest activation pressure we can apply (forced-active initialization plus direct supervision of the router on a regime axis), the modulation **decays monotonically back to identity** on every objective and dataset we test. This is the paper's spine, and it is a statement about *training dynamics*, not about expressivity: a *frozen* router probes to 0.92 agreement with the regime bucket, so the capacity exists; joint training will not use it.

The explanation is a **gauge symmetry of the block parameterization**. An input-side channel-wise scale folds *exactly* into the block's input projections as a right-multiplied diagonal $W_\bullet \operatorname{diag}(\gamma)$, and the shift folds into their biases, with no change to the function the block computes. Identity is therefore a flat, lossless valley—a gauge direction—and the optimizer slides along it, returning (γ, β) to $(\mathbf{1}, \mathbf{0})$. This predicts, and we observe, a decay signature that is a property of the *block*, not the data: a forced-active escalation decays with the **same time constant** $\tau \approx 7000$ **steps (**$R^2 > 0.999$**)** on FI-2010 equities and Kaggle crypto—two markets with nothing in common but the optimizer that trains them.

We then run the test the gauge argument names: an **output-side** placement that gates the block's output. It does not escape absorption either—each Mamba-3 block ends in a *bias-free* output projection, so the output scale folds into *that* (the same gauge move on the block's other face, same τ), while the non-absorbable shift β decays in lockstep, killed by reconstruction dominance. The null is therefore **overdetermined**, leaving the only escape route *inside* the fused selective-scan kernel—the kernel-level change our hardware cannot run.

Why this is a foundation-model question. Although we instantiate the study on a financial world model, the object of study is a property of the Mamba block itself. Gauge absorption is an *algebraic* property of the block's parameterization—a channel-wise affine on the input folds into the same $W_\Delta / W_B / W_C$ projections whether the block sits in a world model, a language model, or any other selective-scan foundation model—so the mechanism, and the design consequence that follows, are **architecture-universal and scale-free** by construction, not specific to finance. The char-level Mamba *language*-model control (§5) plays a narrower role: it maps *where* the boundary lies by exhibiting a strongly-decodable regime on which the modulation engages. It is a boundary-mapping probe, not the bridge the universality argument rests on—that bridge is the algebra.

Contributions.

1. A **decisive, mechanistically explained null** for input-affine FiLM conditioning of a selective scan: the modulation collapses to identity under joint training and cannot be forced active, across five objectives and three LOB distributions (§4).
2. The mechanism—**gauge absorption**—with its dataset-invariant signature ($\tau \approx 7000$, $R^2 > 0.999$) and an **output-side placement that makes the null overdetermined**, locating the boundary *inside the kernel*; **demonstrated**, not just argued, by a toy reference-scan experiment and a no-weight-decay ablation showing the decay is **weight-decay-driven over a flat loss** (§5).
3. A **boundary at regime decodability**: a Mamba LM control (prose vs. code) shows a *strongly*-decodable regime engages the modulation, scoping the collapse to *weakly*-decodable conditioning (§5).
4. A **load-bearing-diagnostic hazard** (the gap reads positive while the mechanism is inert) and a **demonstration substrate** with an architecture-agnosticism check and a deflated, architecture-independent economic coda (§6).

The financial machinery (a Mamba-3 world model in the DreamerV3/Drama lineage) is the stress-substrate that let us probe the mechanism on genuinely different data; it is described only to the depth the null requires.

2 Related Work

Selective state-space models and Mamba. S4, S5, and Mamba give $O(n)$-complexity alternatives to attention (Gu et al., 2022; Smith et al., 2023; Gu & Dao, 2024). Mamba (Gu & Dao, 2024) introduced *selective* state spaces in which Δ and $\mathbf{B}$ are input-dependent, enabling content-aware filtering; Mamba-2 (Dao & Gu, 2024) cast this as structured masked attention; Mamba-3 (Lahoti et al., 2026) adds MIMO state transitions. Our object of study is not a new SSM but *how a selective scan trains under an added affine*—that the very input-dependence which makes the modulation seem natural is what makes it gauge-absorbable.

FiLM and conditioning. Feature-wise linear modulation (Perez et al., 2018) injects context via per-channel affine parameters and is a default conditioning primitive across vision and multi-task learning. We are not aware of prior work that asks whether a FiLM affine *persists* under joint training of a selective scan, or that identifies its absorption into the host projections as a gauge direction.

Scale redundancy and scale-invariance. That a channel-wise affine placed before a linear map is redundant with that map is a known theme in the normalization literature. van Laarhoven (2017) shows that weights feeding a normalization layer are scale-invariant, so L_2 regularization acts on their *scale*—and thus the effective learning rate—rather than as a capacity control; Hoffer et al. (2018) and Li & Arora (2020) build on the same scale-invariance to re-derive weight decay as an effective-learning-rate schedule. Our null is an instance of this redundancy in a setting it has not been analyzed: the absorbed affine is not a static normalization gain but a per-regime, *input-dependent* FiLM folding into the selective scan's own $W_\Delta/W_B/W_C$ projections. What is new is the input-dependent selective-scan instantiation, the dataset-invariant decay signature that attributes the collapse to the optimizer, the output-side placement showing the null is *overdetermined*, and the practical consequence that the only kernel-friendly, drop-in form is exactly the absorbable one—the default a practitioner reaches for, not a corner case.

World models and the LOB substrate. DreamerV3 (Hafner et al., 2023) established the categorical-latent RSSM template and Drama (Wang et al., 2024) replaced its recurrence with a Mamba-2 selective scan; we instantiate this lineage with a Mamba-3 backbone purely as a substrate (no spine result depends on the world-model framing beyond a reconstruction objective and three distinct distributions). For substrate context, DeepLOB (Zhang et al., 2019) and the FI-2010 benchmark (Ntakaris et al., 2018) anchor LOB direction prediction, the LOBCAST benchmark (Prata et al., 2023) documents the in-distribution macro-F1 range confirming our data carries signal, and classical regime-switching (Hamilton, 1989) and changepoint detection (Adams & MacKay, 2007) motivate a regime axis; we use the Kaufman *efficiency-ratio* predictability axis as the most defensible regime signal in the study.

3 Background: the selective scan and the FiLM modulator

Selective state-space model. A discretized SSM maps x_t to y_t through a latent state, $h_t = \bar{\mathbf{A}} h_{t-1} + \bar{\mathbf{B}} x_t$, $y_t = \mathbf{C} h_t$. Mamba makes the scan parameters **input-dependent** via learned linear projections of the block input x_t:

$$\Delta_t = \sigma(W_\Delta\, x_t), \qquad \mathbf{B}_t = W_B\, x_t, \qquad \mathbf{C}_t = W_C\, x_t. \tag{1}$$

This selection mechanism is exactly the property a regime-conditioner would exploit: *any* affine transformation of x_t propagates into all three scan parameters. It is also, as §5 shows, exactly the property that makes an input affine absorbable.

RegimeFiLM modulator. A regime head infers a soft categorical distribution over R regimes (the regime count: $R = 4$ on the FI-2010/Kaggle spot books, $R = 8$ on Polymarket) from the stem output and reads out a low-dimensional regime embedding r_t; a hypernetwork maps r_t to per-block FiLM parameters, bounded for bf16 stability:

$$\boldsymbol{\alpha}_t = \mathrm{softmax}(W_r\, x_t^{(0)}) \in \mathbb{R}^R, \quad r_t = \boldsymbol{\alpha}_t^\top E, \quad \gamma^{(\ell)} = 1 + \tanh(\tilde{\gamma}^{(\ell)}), \;\; \beta^{(\ell)} = \tanh(\tilde{\beta}^{(\ell)}). \tag{2}$$

Family	Objective	Dataset	Gap (sign, unit)	film_g	Verdict
Volatility/scale	Student-t NLL	FI-2010	≈ 0 nats	$\approx$ ident.	NULL
Order-flow volume	Volume NLL	FI-2010	≈ 0 nats	$\approx$ ident.	NULL
Direction	macro-F1	FI-2010	$+0.008^{\dagger}$ F1	0.018	NULL
Settlement	BCE log-loss	Polymarket	$+0.18^{\dagger}$ nats	0.008	NULL
PnL	simulated trade	Polymarket	$-\$627^{\dagger}$	0.010	NULL

Table 1: RegimeFiLM is inert across every objective. reg_H equals its uniform maximum ($\log R$) in all rows. The "Gap" column is in each objective's *own* units (reconstruction NLL nats, macro-F1 points, BCE log-loss nats, dollars), so its magnitude is not comparable across rows; only its sign and the inert film_g are. † a non-zero gap whose sign is a distributional artifact, not a FiLM effect (FiLM is inert, and the treatment is absolutely worse). The PnL row's $-\$627$ is the FiLM A/B gap on the PnL *objective* (treatment$-$control), *not* an economic total (cf. the $+\$5{,}147$ settlement total in §6).

The hypernetwork is initialized to zero, so $\gamma^{(\ell)} = \mathbf{1}$ and $\beta^{(\ell)} = \mathbf{0}$ at step 0—the modulated backbone is **exactly the unmodulated baseline at initialization**, an exact paired control. A Switch-Transformer-style load-balance term (Fedus et al., 2022) maximizes the entropy of the *batch-averaged* router distribution, $\mathcal{L}_{\text{regime}} = \lambda_{\text{ent}}(\log R - H[\bar{\boldsymbol{\alpha}}])$, regularizing marginal regime usage while leaving per-step assignments free to be peaky. (We ablate this term away in the strongest-null test, below, so the observed router uniformity is not attributable to it.)

Two application sites. The **input-side** placement routes the regime into $\Delta_t, \mathbf{B}_t, \mathbf{C}_t$ through the selection mechanism:

$$x_t^{(\ell)} = x_t^{(\ell-1)} + \mathrm{Mamba3}\Big(\gamma^{(\ell)} \odot \mathrm{RMSNorm}(x_t^{(\ell-1)}) + \beta^{(\ell)}\Big). \qquad \text{(input-side)}$$

As a control we also expose a **post-scan** site that gates the block *output*, sharing the identical modulator and differing only in where the affine lands:

$$x_t^{(\ell)} = x_t^{(\ell-1)} + \gamma^{(\ell)} \odot \mathrm{Mamba3}\Big(\mathrm{RMSNorm}(x_t^{(\ell-1)})\Big) + \beta^{(\ell)}. \qquad \text{(post-scan)}$$

The input-side scale folds into $W_\Delta/W_B/W_C$; the post-scan scale folds into the bias-free output projection W_{out}; only the post-scan additive shift is non-absorbable (§5).

Diagnostics. We gate every claim on activation diagnostics: $\mathrm{film}_g = \text{mean}\,|\gamma - 1|$ (deviation of the multiplicative gate from identity, zero at identity), its additive counterpart $\mathrm{film}_b = \text{mean}\,|\beta|$ (which we report for the output-side shift in §5), and the router entropy reg_H (whose uniform maximum is $\log R$). A positive generalization gap with an inert FiLM is an artifact, not a regime-conditioning effect.

4 RegimeFiLM is a decisive null

The result. Across five objectives—Student-t NLL (volatility/scale), volume NLL[1] (order-flow), direction macro-F1, Polymarket settlement BCE, and a simulated-trading PnL objective—FiLM collapses to identity under joint training. In every family, on both datasets and on a money objective, film_g decays toward zero and the router entropy reg_H sits at its uniform maximum ($\log R$: $\log 4$ on FI-2010, $\log 8$ on Polymarket). Where a generalization gap reads positive (direction $+0.008$, settlement $+0.18$, by sign), FiLM is provably inert, the treatment is *absolutely worse* in both regimes, and the gap is a high$-$low distributional artifact whose sign flips with head calibration—which is exactly why the activation diagnostics are load-bearing.

[1]The volume-NLL row re-evaluates the Student-t checkpoints on the order-flow channel—a fifth *objective*, not an independently trained run.

Dataset	Asset / resolution	Gap (pred / vol)	film_g	Verdict
FI-2010	event stream	≈ 0 / ≈ 0	0.004	NULL
Kaggle BTC	1 min	≈ 0 / ≈ 0	0.004	NULL
Kaggle ETH	1 min	≈ 0	0.005	NULL
Kaggle ADA	1 min	≈ 0	0.005	NULL
Kaggle BTC	5 min	≈ 0	0.005	NULL
Kaggle BTC	1 s (slice)	≈ 0	0.005	NULL

Table 2: Cross-dataset confirmation of the regime-FiLM null. $\text{reg}_H = \log 4$ (uniform router) in every row. The null is dataset-, asset-, and resolution-independent.

Strongest-null escalation. We force FiLM active (raising the init so $\text{film}_g = 0.150$ at step 0) and supervise the router *directly* on the spot efficiency-ratio axis with the strongest settings (LR multiplier 50, supervision weight 30, decoupled router/FiLM, fed observation volatility). The modulation still decays monotonically toward identity—$\text{film}_g : 0.150 \to 0.119 \to 0.100 \to 0.074 \to 0.056 \to 0.043$, still falling—with $\text{reg}_H = \log 8$ throughout. The capacity to use the regime is present but goes unused: a frozen router probes to 0.92 agreement on the *volatility* bucket—the axis the representation encodes best—yet even there the modulation collapses, while on the *predictability* axis this escalation actually supervises the regime decodes only weakly (≈ 0.25–0.36). The capacity that exists is thus not even the one joint training declines to use. This is the strongest form of the null: the optimizer drives the modulation to identity regardless of objective or forced supervision, for the weakly-decodable predictability and volatility regimes we study—a language-model control in §5 maps where this boundary lies. (This forced-active run starts the gate at $\text{film}_g = 0.150$; the cross-dataset escalation below uses a stronger 0.250 initialization—a *different* run, not a continuation.)

Cross-dataset confirmation and the τ-invariant signature. We take the finding to two further LOB settings—the Kaggle high-frequency crypto spot book and FI-2010 re-examined under a *predictability*-primary regime axis (efficiency-ratio window split; realized volatility secondary)—judged by forecasting metrics alone. Across FI-2010 and the Kaggle book (BTC, ETH, ADA at 1 s/1 min/5 min), the zero-initialized modulation never leaves identity: $\text{film}_g \approx 0.004$ and $\text{reg}_H = \log 4$ in every setting (Table 2), and with FiLM an exact paired control every held-out gap is statistically indistinguishable from zero (all 95% CIs include 0). The split is non-degenerate—reference windows carry more structure on all three predictability measures (efficiency ratio, permutation entropy, Hurst), and DeepLOB attains 0.628 macro-F1 on the FI-2010 horizon-10 labels—so the null is a negative on forecastable data, not a no-signal artifact.

The collapse is optimizer-driven, not data-specific. Run as a forced-active, ER-supervised escalation (starting at $\text{film}_g = 0.250$), the modulation decays monotonically at every logged step on *both* datasets ($0.250 \to 0.144$ Kaggle, $0.250 \to 0.147$ FI-2010), with $\text{reg}_H \approx \log 4$ throughout **even though the load-balance term is disabled**—so the uniform router is not a load-balance artifact. An exponential fit $a\, e^{-t/\tau} + c$ matches both trajectories at $R^2 > 0.999$ with near-identical parameters ($a \approx 0.30$, $\tau \approx 7000$ steps). The decay is monotone with a negative slope sustained to the last logged step—where film_g is still ≈ 0.15 and falling—and the fitted asymptote lies at or below identity ($c \lesssim -0.05$, growing *more* negative when the lower bound on c is relaxed), so the modulation extrapolates toward identity rather than to a positive plateau—we rest the claim on the observed monotone decay, with exact identity as the extrapolated limit. That the decay rate is essentially **identical** across two very different markets points to the optimizer, not the data, as the driver (Figure 2).

5 Mechanism: gauge absorption, and an overdetermined null

Why identity is a free direction. The decay has a simple mechanistic explanation that also *bounds* the claim. The input-side modulation $\tilde{x} = \gamma \odot \hat{x} + \beta$ is applied to a block

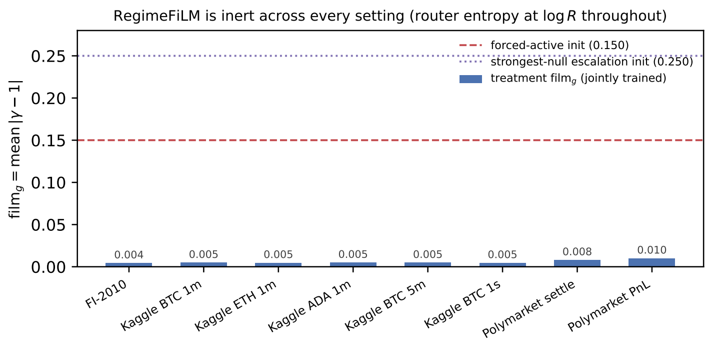

Figure 1: **RegimeFiLM is inert across every A/B setting.** The jointly-trained treatment $\text{film}_g = \text{mean}\,|\gamma - 1|$ (bars) sits near 0.004–0.010 on every dataset, asset, and resolution—FI-2010, Kaggle BTC/ETH/ADA at three bar widths, and the two Polymarket objectives—while a forced-active initialization starts at 0.150 and the strongest-null escalation at 0.250 (dashed lines). The router entropy sits at its uniform maximum $\log R$ throughout. The modulation never grows from its exact identity initialization.

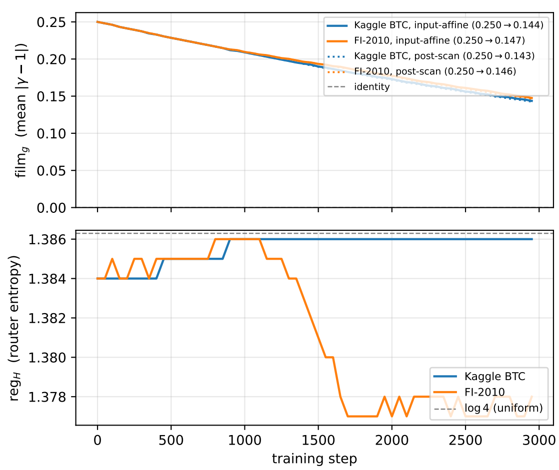

Figure 2: **Strongest-null test and its placement-invariance.** Forcing the modulation active ($\text{film}_g = 0.250$ at step 0) and supervising the router directly on the efficiency-ratio bucket still decays film_g monotonically toward identity on both datasets (top, solid), with reg_H pinned at its uniform maximum $\log 4$ throughout (bottom). The exponential fit gives $R^2 > 0.999$ with a dataset-invariant time constant $\tau \approx 7000$ and a negative fitted asymptote, so the modulation extrapolates toward identity rather than to a positive plateau. The **post-scan** (output-side) gate (top, dotted) decays along the same trajectory—its scale folds into the bias-free output projection just as the input-side scale folds into $W_\Delta/W_B/W_C$—so the null is invariant to where the affine is applied (§5).

input that the very next operations re-project: $\Delta_t = \sigma(W_\Delta \tilde{x}_t)$, $\mathbf{B}_t = W_B \tilde{x}_t$, $\mathbf{C}_t = W_C \tilde{x}_t$. A constant channel-wise scale folds **exactly** into those projections as a right-multiplied diagonal $W_\bullet\,\text{diag}(\gamma)$, and the shift folds into their biases, with **no change to the function the block computes**; the host weights can thus represent any constant (γ, β) themselves. Identity is therefore a **gauge** direction—a flat, lossless valley the optimizer can slide along—so joint training folds the modulation into the host weights and returns (γ, β) to $(\mathbf{1}, \mathbf{0})$, which is precisely the monotone decay we observe. Among the equivalent points on this flat orbit, weight decay on the bounded modulator selects the minimum-norm one ($\gamma{=}\mathbf{1}$, $\beta{=}\mathbf{0}$, its zero-init) while the host weights absorb the scale at no loss cost—so the descent to identity is driven by weight decay on the modulator, not by the objective, a prediction the weight-decay control below confirms. The folding identity is exact for the *constant* part of the affine, whereas the trained modulator emits an *input-dependent* (γ, β); the causal order that closes the argument is therefore that on a weakly-decodable regime the supervised router collapses to uniform ($\mathrm{reg}_H = \log R$), the modulation reduces to its constant part, and weight decay folds *that* part to identity. The live, input-dependent mechanism reduces to the gauge one precisely when the regime is hard to decode—which the language-model control below confirms by exhibiting the opposite outcome when it is not.

This sharpens the result two ways. First, it explains the **dataset-invariant decay constant** (Figure 2): the absorption direction is a property of the block parameterization, not of the data. Second, it **scopes** the null: the claim is that *input-affine* conditioning of a selective scan is gauge-absorbable and will not persist under joint training—not that regime conditioning is impossible in principle. Mechanisms outside this gauge—modulating $W_\Delta / W_B / W_C$ directly *inside* the kernel, or conditioning a quantity inserted *between* the block's bounding projections—are not trivially absorbable and are the natural next test. Our contribution is the negative that matters in practice: the most kernel-friendly, drop-in form of the idea, the one that needs no change to the fused CUDA kernel, is exactly the form the optimizer can and does undo.

An output-side placement: the null is overdetermined. The gauge argument names an obvious next test—gating *after* the scan—so we run it, under the *identical* forced-active, ER-supervised escalation, changing only where the learned (γ, β) are applied. Instead of the input-side $\tilde{x} = \gamma \odot \hat{x} + \beta$, we gate the block output $y' = \gamma \odot y + \beta$. This placement does **not** escape absorption, and the reason is instructive. Each Mamba-3 block ends in a **bias-free** linear output projection $y = W_{\mathrm{out}}\, h$, so a channel-wise output scale folds exactly into it, $\gamma \odot (W_{\mathrm{out}} h) = (\mathrm{diag}(\gamma)\, W_{\mathrm{out}})\, h$—the same gauge move as the input side, now on the block's *other* face. The additive shift β is different: with no output bias to absorb it and a nonlinear RMSNorm separating it from the next block's projections, β is genuinely **non-absorbable**.

The escalation resolves both at once. Under forced activation and direct ER supervision, film_g *and* film_b decay monotonically and in lockstep on both datasets ($\mathrm{film}_g : 0.250 \to 0.147$ FI-2010, $0.250 \to 0.143$ Kaggle; film_b tracks it within 0.005), with $\mathrm{reg}_H \approx \log 4$ throughout. The exponential fit gives the **same** dataset-invariant signature as the input-side gate—$a \approx 0.30$, $\tau \approx 7065$ (FI-2010) / 6598 (Kaggle), a negative fitted asymptote ($c \lesssim -0.05$, likewise below identity with the bound relaxed), $R^2 > 0.999$—so the same monotone decay toward identity holds here too.

The null is therefore **overdetermined**: no component is left with a surviving direction, by two routes that share a single driver—weight decay acting on a loss that rewards neither.

- **Gauge absorption** disposes of every direction the host weights can represent—the multiplicative scale on *either* face of the block—which is why the decay constant is invariant to placement.
- **Reconstruction dominance** disposes of the one direction they cannot: the non-absorbable shift β, returned to identity at the same rate purely because it earns nothing under the reconstruction objective.

The two are thus not independent causes so much as two faces of the same unrewarded-direction-under-weight-decay mechanism—an absorbable scale and a non-absorbable but unrewarded shift—which is why β decays in lockstep with γ at a shared rate rather than on its own schedule. A modulation escaping both would have to act *between* the block's bounding projections—modulating $W_\Delta / W_B / W_C$ within the fused selective-scan kernel—which is precisely the kernel-level change outside our hardware envelope (§6). We thus delineate the boundary not as a parameterization that survives training, but as the location—inside the kernel—a surviving modulation would have to occupy.

The boundary, demonstrated. We make this concrete at toy scale on the pure-PyTorch *reference* scan (the slow path `mamba_ssm` ships, which the fused-kernel hardware limit does not touch). An input-affine modulation folds into the host projections to floating-point precision (max output deviation 1.2×10^{-6}) and **decays to identity** under joint training with weight decay ($\text{film}_g : 0.086 \to 0.010$), while the *same* modulation applied to the projected $\Delta/\mathbf{B}/\mathbf{C}$ inside the scan is provably non-absorbable (best-fit fold residual 1.3×10^{-2}, $\sim 10^4\times$ larger) and **persists** ($\text{film}_g : 0.093 \to 0.175$, seeds 0–3). A weight-decay control isolates the driver: with weight decay removed the full-model escalation no longer decays—the time constant blows up from $\tau \approx 6685$ ($R^2 = 0.9999$) to $\tau \approx 3 \times 10^5$ ($R^2 = 0.48$, no clean decay)—and the toy reproduces the same flattening (input-affine $0.086 \to 0.342$ with weight decay off vs. $0.086 \to 0.010$ with it on). The gauge-direction decay is therefore **driven by weight decay over a flat loss**, not by the objective. The boundary is thus demonstrated, not asserted: the in-kernel placement is non-absorbable and survives the very training that drives the affine form to identity.

The regime is decodable yet unused. Two probes localize the failure. The jointly-supervised router is uniform *per step* (per-sample entropy $\approx \log 4$, near-chance ER agreement 0.25–0.31): supervision does not even make it regime-discriminative. A frozen linear probe decodes the predictability bucket only weakly (0.33–0.36 vs. 0.25) and volatility better (0.43–0.47)—yet the volatility-axis gap is *also* null. So even on the best-encoded axis, and even with the regime supplied by supervision, the in-scan modulation collapses to identity.

Language-model control: the null is bounded by regime decodability. Because the collapse is optimizer-driven, the natural worry is that it is nonetheless specific to LOB data; a minimal language-model control maps where the boundary lies. We train a 2-layer char-level Mamba LM (built directly on `mamba_ssm` blocks) on a two-domain corpus—English prose vs. Python source—with the per-position regime inferred by a context-aware router supervised on the domain and the *same* input-affine FiLM gating the second block. Here the regime is *strongly* decodable: the router reaches ~ 0.95 domain accuracy, and the FiLM does **not** collapse—film_g rises from 0.11 to 0.52 and plateaus, the opposite of the LOB decay. This sharpens the mechanism rather than counters it. Gauge absorption removes the *constant* component of the affine unconditionally (the folding test above); the *input-dependent* component survives only when the regime is decodable enough to be worth encoding. In every LOB setting the predictability/volatility regime is weakly decodable and the router collapses to uniform ($\text{reg}_H = \log R$), so the modulation reduces to its gauge-absorbable constant part and decays; a strongly-decodable regime engages it. We therefore scope the claim accordingly: the optimizer-driven collapse holds for **weakly-decodable** regime conditioning of selective scans—the predictability and volatility axes a financial world model actually faces—not for all conditioning. (Single-seed toy; the router$\Rightarrow$persistence outcome is clear, the plateau value is not load-bearing.)

The gap is real, and FiLM closes none of it. The null is not an absence of anything to fix: on the Kaggle book reconstruction NLL is $+0.105$ worse on high- than low-volatility windows in *both* arms, yet RegimeFiLM narrows the degradation by only ~ 0.001 ($\sim 1\%$) while film_g is inert—it fails to engage a gap that is plainly present.

6 Demonstration substrate and architecture-agnosticism

The mechanism above is the subject; the financial machinery is the **substrate** that let us probe it on three genuinely different distributions, compressed to what the null and its agnosticism check require.

Substrate. We build a Mamba-3 world model for LOB microstructure in the DreamerV3/Drama lineage: a Transformer-over-depth encoder feeds a stack of Mamba-3 SISO blocks driving a categorical-latent dynamics prior under a reconstruction objective, with only the settlement and book-relative-EV heads bearing on any result below (the rest of the machinery is off the spine). For Polymarket we expose the **spot path** that *defines* settlement (the underlying Chainlink close vs. open reference) plus a time-to-expiry clock and TTE-weighted target—the causal signal of which the settlement label is a function.

Backbone	Params	FI-2010 recon NLL	Kaggle recon NLL
Mamba-1	11.45M	$-\mathbf{0.5403}$	0.0241
Mamba-2	10.20M	-0.5309	$-\mathbf{0.0213}$
Mamba-3 SISO	10.30M	-0.5346	0.0207
Transformer (post-norm)	9.90M	-0.5175	0.0191
Transformer (pre-norm)	9.61M	-0.5041	0.0224

Table 3: Matched-parameter backbone ablation (held-out reconstruction NLL, lower is better). No backbone wins both datasets—Mamba-1 leads on FI-2010, Mamba-2 on Kaggle—so the choice of sequence model is not the operative variable.

Backbone ablation (no backbone dominates). Under matched parameter budgets ($\sim$ 10.3M; actuals in Table 3), seed 0, 3000 steps, on FI-2010 and the Kaggle BTC spot book, we report held-out Student-t reconstruction NLL. The result is a genuine ambiguity, not a clean win: **no backbone leads on both datasets**, and a pre-norm Transformer matched to the Mamba-3 block (9.61M) is likewise competitive but non-dominant (-0.5041 / 0.0224)—not an artifact of an under-tuned post-norm baseline. This is exactly what a result about how selective scans *train*, rather than which one is best, should predict.

We run Mamba-3 SISO throughout: the MIMO kernel's dynamic shared-memory footprint exceeds our GPU cap at every warp-valid (rank, chunk) configuration, so MIMO is outside our hardware envelope—the same kernel-level region the null's escape route would occupy (§5).

The economic coda is architecture-independent (deflated). Given the spot path, the model's directional settlement probability—traded by a selective-participation strategy that bets only when a train-frozen predictability (efficiency-ratio) gate fires, summarized by a **market-block** bootstrap (whole markets resampled, so correlated trades count once)—is a survivable edge in the regime the gate targets ($\mathrm{P}(\text{profit}) \approx 1.0$, 95% CI [+\$2,445, +\$7,747] over 127 markets). Crucially the edge is **architecture-independent**: a logistic regression on the *same* spot-conditioned features reproduces and exceeds it (+\$6,914 vs. the settlement head's +\$5,147). Binary settlement is near-deterministic in the spot move since open, so a linear readout captures it as cleanly as the sequence model—the operative variable is the features and the gate, **not** the backbone, confirming the null's lesson from the other side. Only the predictability gate is confirmatory (threshold frozen before held-out evaluation); the depth-band/early-market cuts, the direct tradeable-EV head that makes the marginal low-predictability regime robust, and the recorded-vintage reliability diagnostic are deferred to the long-form version.

Limitations. The *null* is multi-setting and multi-seed; the *constructive* claims that bound it are smaller. The optimizer attribution rests on a τ matched across two datasets plus a within-model weight-decay ablation; the escalation fits are seed-0 (the seeds-0–3 in-kernel toy supplies the multi-seed check); the language-model control is a single-seed 2-layer toy; the in-kernel "survives" result is at toy scale; and the substrate is finance-only, so the gauge argument's architecture-universality is algebraic, not demonstrated at FM scale. None of these bears on the null.

7 Conclusion

We studied the most natural, kernel-friendly way to make a Mamba backbone regime-aware—an input-affine FiLM on each block—and found a **decisive training-dynamics null**: the modulation collapses to identity under joint training and cannot be forced active even by direct supervision. The cause is a **gauge symmetry**—the affine folds losslessly into the block's own $W_\Delta/W_B/W_C$ projections, so the optimizer slides it back to identity ($\tau \approx 7000$, $R^2 > 0.999$); an **output-side placement makes the null overdetermined**, leaving the only escape route *inside* the fused kernel. The upshot is architecture-universal: only a placement *between* a block's bounding projections—inside the kernel—can persist.

References

R. P. Adams and D. J. C. MacKay. Bayesian online changepoint detection. *arXiv preprint arXiv:0710.3742*, 2007.

T. Dao and A. Gu. Transformers are SSMs: Generalized models and efficient algorithms through structured state space duality. In *Proc. ICML*, 2024.

W. Fedus, B. Zoph, and N. Shazeer. Switch Transformers: Scaling to trillion parameter models with simple and efficient sparsity. *JMLR*, 23:1–39, 2022.

A. Gu, K. Goel, and C. Ré. Efficiently modeling long sequences with structured state spaces. In *Proc. ICLR*, 2022.

A. Gu and T. Dao. Mamba: Linear-time sequence modeling with selective state spaces. In *Proc. COLM*, 2024.

D. Hafner, J. Pasukonis, J. Ba, and T. Lillicrap. Mastering diverse domains through world models. *arXiv preprint arXiv:2301.04104*, 2023.

J. D. Hamilton. A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2):357–384, 1989.

E. Hoffer, R. Banner, I. Golan, and D. Soudry. Norm matters: Efficient and accurate normalization schemes in deep networks. In *Proc. NeurIPS*, 2018 (arXiv:1803.01814).

A. Lahoti, K. Y. Li, B. Chen, C. Wang, A. Bick, J. Z. Kolter, T. Dao, and A. Gu. Mamba-3: Improved sequence modeling using state space principles. In *Proc. ICLR*, 2026 (arXiv:2603.15569).

Z. Li and S. Arora. An exponential learning rate schedule for deep learning. In *Proc. ICLR*, 2020 (arXiv:1910.07454).

A. Ntakaris, M. Magris, J. Kanniainen, and M. Gabbouj. Benchmark dataset for mid-price forecasting of limit order book data with machine learning methods. *Journal of Forecasting*, 37(8):852–866, 2018.

E. Perez, F. Strub, H. de Vries, V. Dumoulin, and A. Courville. FiLM: Visual reasoning with a general conditioning layer. In *Proc. AAAI*, 2018.

M. Prata, G. Masi, L. Berti, V. Arrigoni, A. Coletta, I. Cannistraci, S. Vyetrenko, P. Velardi, and N. Bartolini. LOB-based deep learning models for stock price trend prediction: A benchmark study. *arXiv preprint arXiv:2308.01915*, 2023.

J. T. H. Smith, A. Warrington, and S. W. Linderman. Simplified state space layers for sequence modeling. In *Proc. ICLR*, 2023.

T. van Laarhoven. L2 regularization versus batch and weight normalization. *arXiv preprint arXiv:1706.05350*, 2017.

W. Wang et al. Drama: Mamba-enabled model-based reinforcement learning is sample and parameter efficient. In *Proc. ICLR*, 2025 (arXiv:2410.08893, 2024).

Z. Zhang, S. Zohren, and S. Roberts. DeepLOB: Deep convolutional neural networks for limit order books. *IEEE Transactions on Signal Processing*, 67(11):3001–3012, 2019.